
Collections Framework (Java)

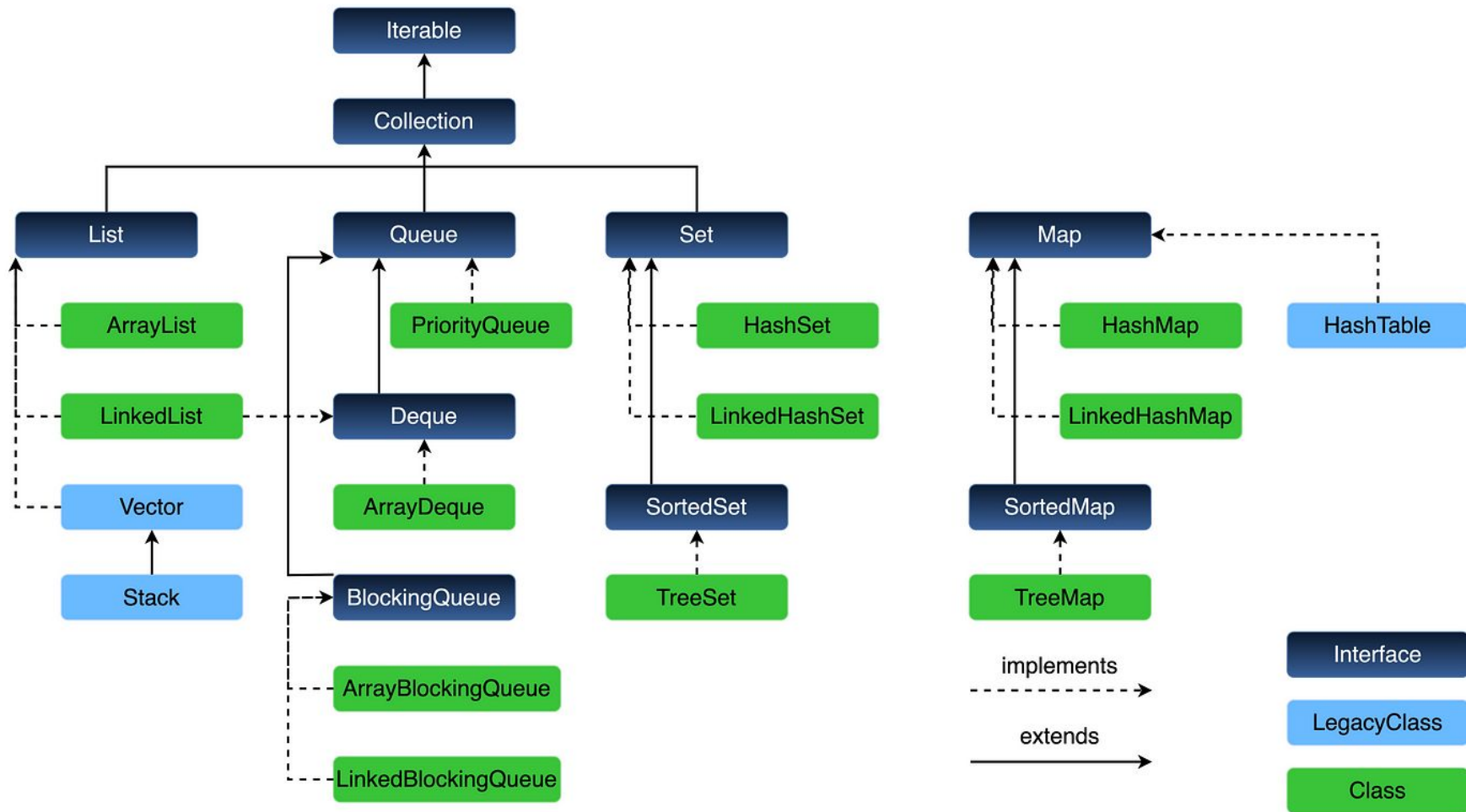
Agenda

01

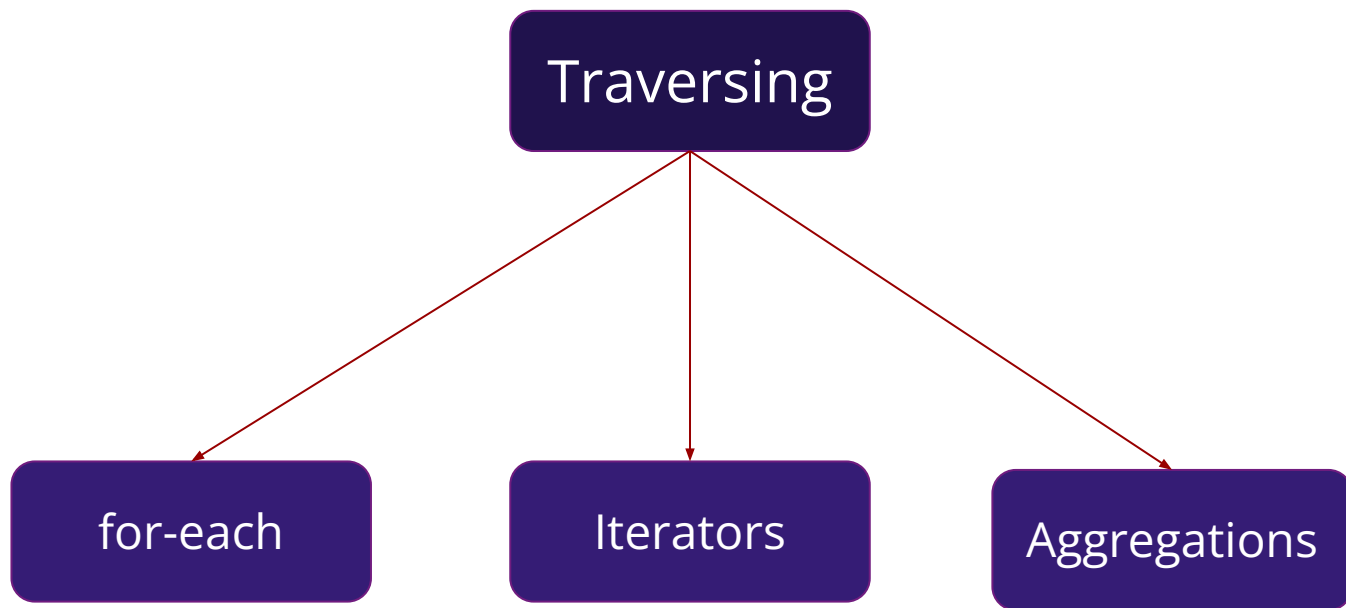
Object Oriented Programming

- Collection Framework
- Iteration
- List
- Queue
- Set

Collections Framework



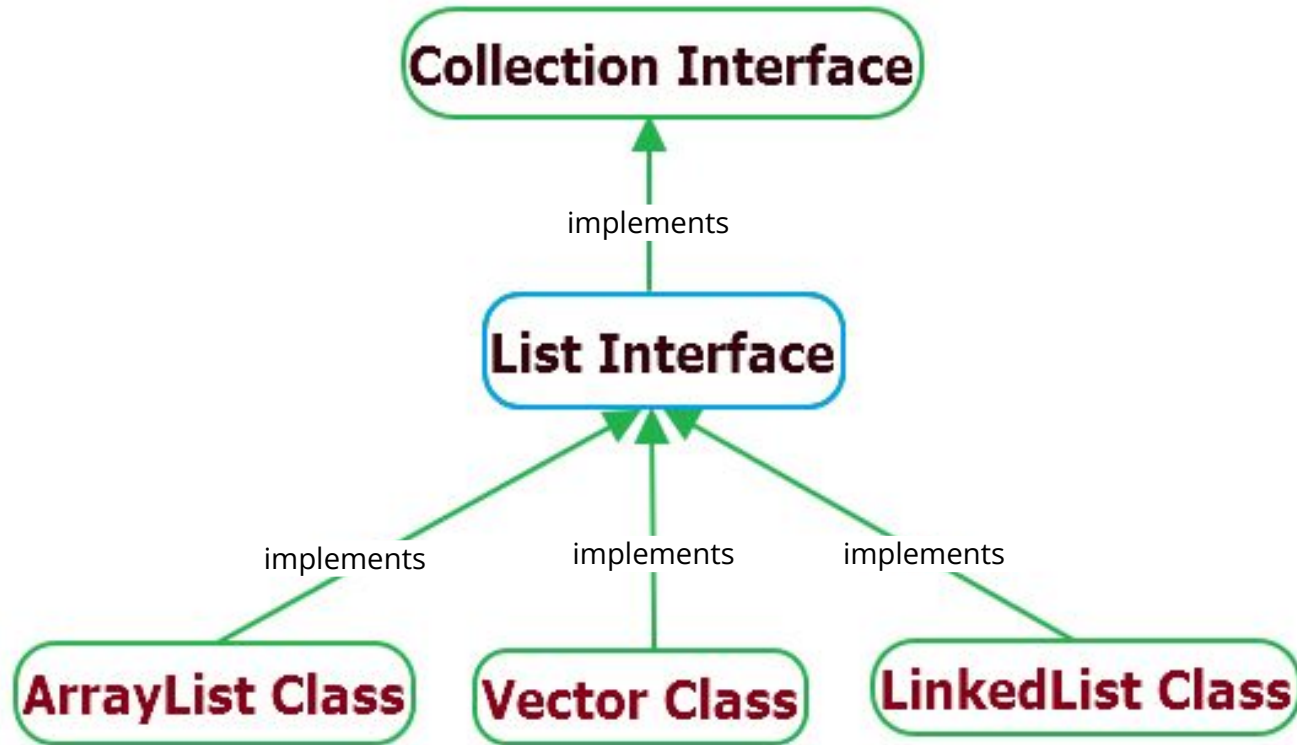
Collections Framework - Traversing



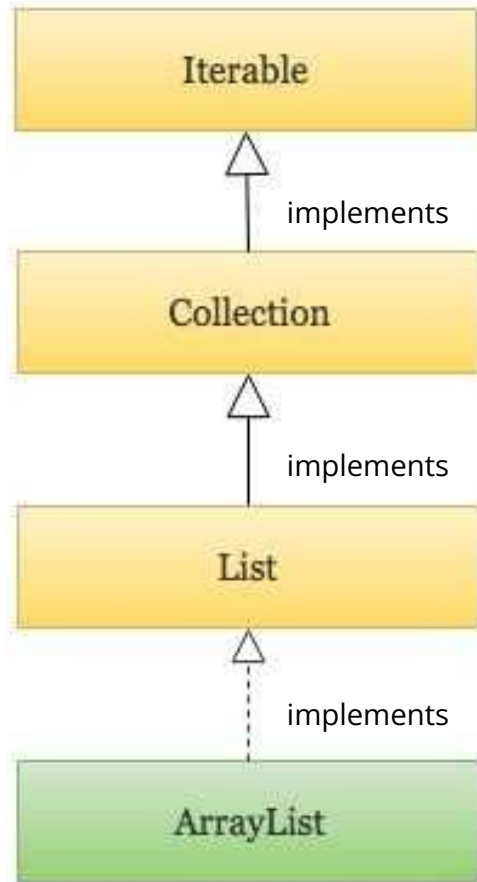
Collections Framework - Comparable - Sorting order

Class	Natural Ordering
Byte	Signed numerical
Character	Unsigned numerical
Long	Signed numerical
Integer	Signed numerical
Short	Signed numerical
Double	Signed numerical
Float	Signed numerical
BigInteger	Signed numerical
BigDecimal	Signed numerical
Boolean	Boolean.FALSE < Boolean.TRUE
File	System-dependent lexicographic on path name
String	Lexicographic
Date	Chronological
CollationKey	Locale-specific lexicographic

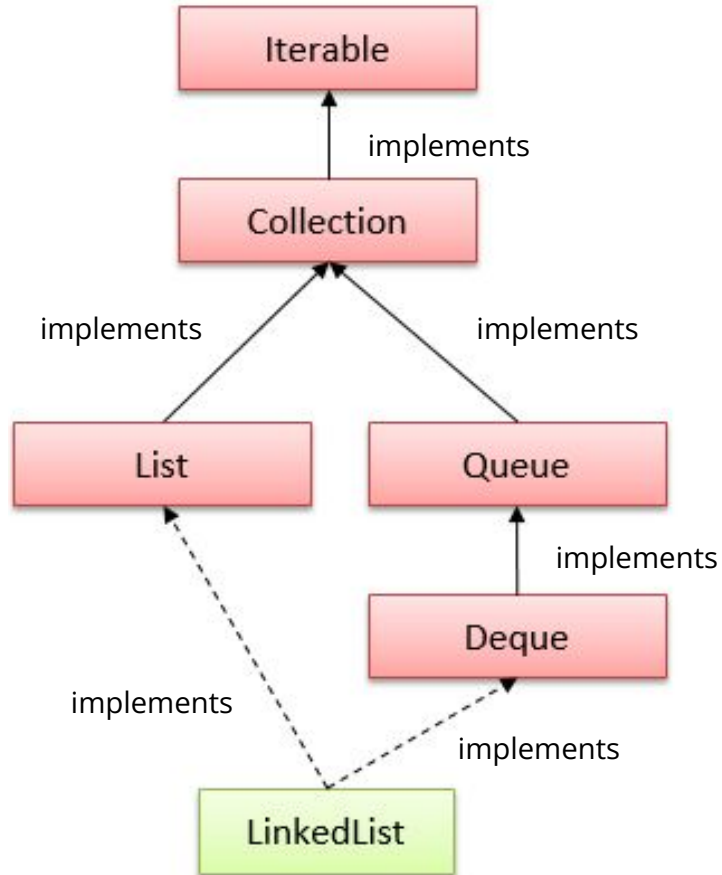
Collections Framework - List



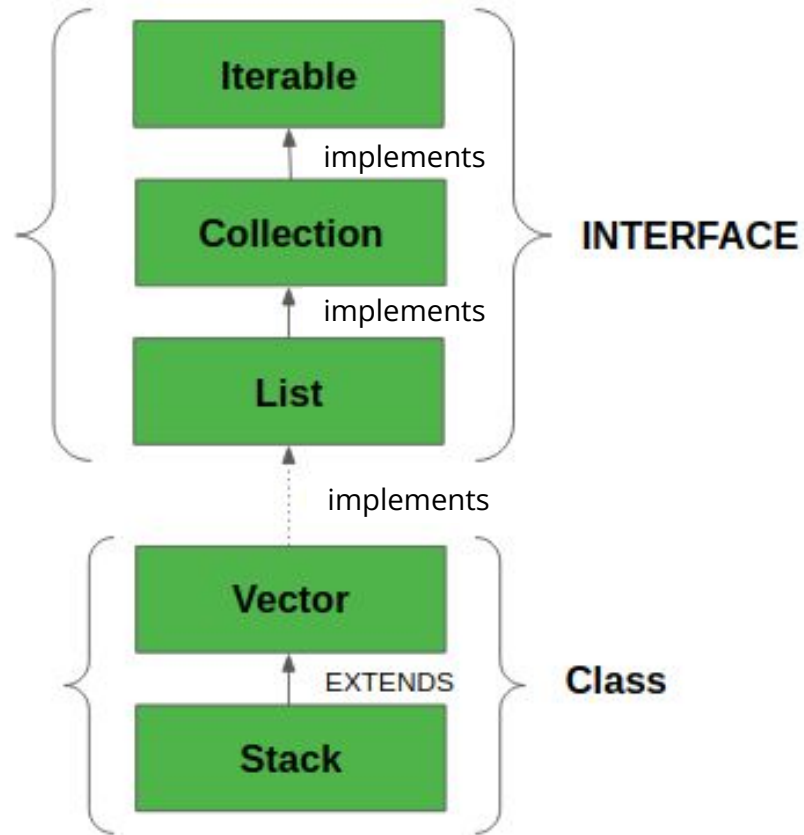
Collections Framework - ArrayList



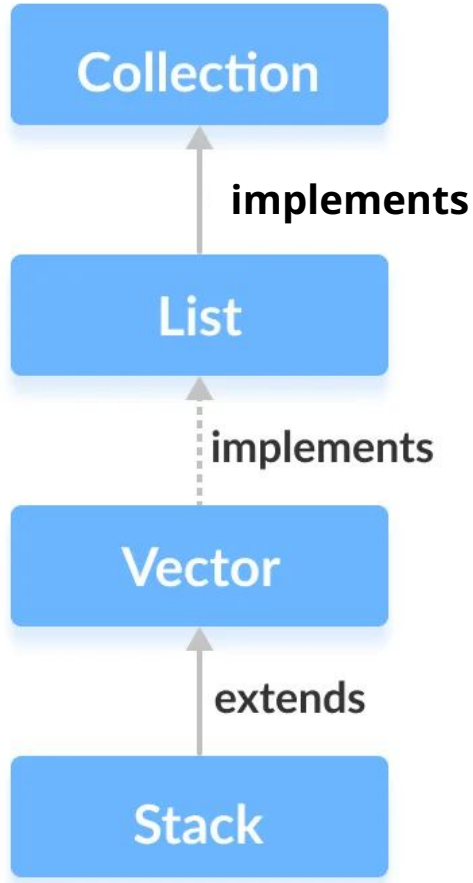
Collections Framework - LinkedList



Collections Framework - Vector



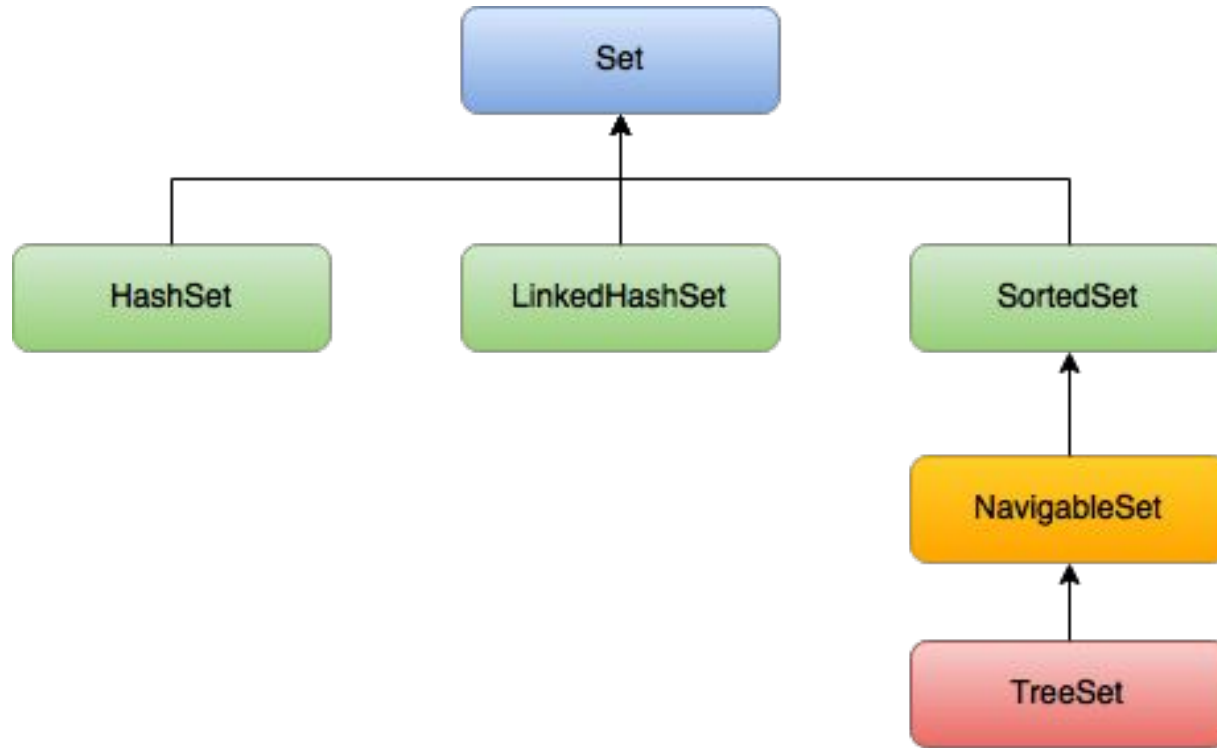
Collections Framework - Stack



Collections Framework - List

Feature / Class	ArrayList	LinkedList	Vector	Stack
Data Structure	Resizable array	Doubly linked list	Resizable array	Extends Vector (Stack behavior)
Access Type	Random access (fast get/set)	Sequential access	Random access	Sequential access (LIFO)
Thread Safety	Not synchronized	Not synchronized	Synchronized (Thread-safe)	Synchronized (Thread-safe)
Legacy	No	No	Yes (older, legacy collection)	Yes (legacy class)
Implements Interface	List	List, Queue, Deque	List	Extends Vector (implements List)
Performance	Fast get/set; slow add/remove in middle	Fast add/remove at ends; slower random access	Similar to ArrayList but slower due to synchronization	Stack-specific methods (push/pop/peek)

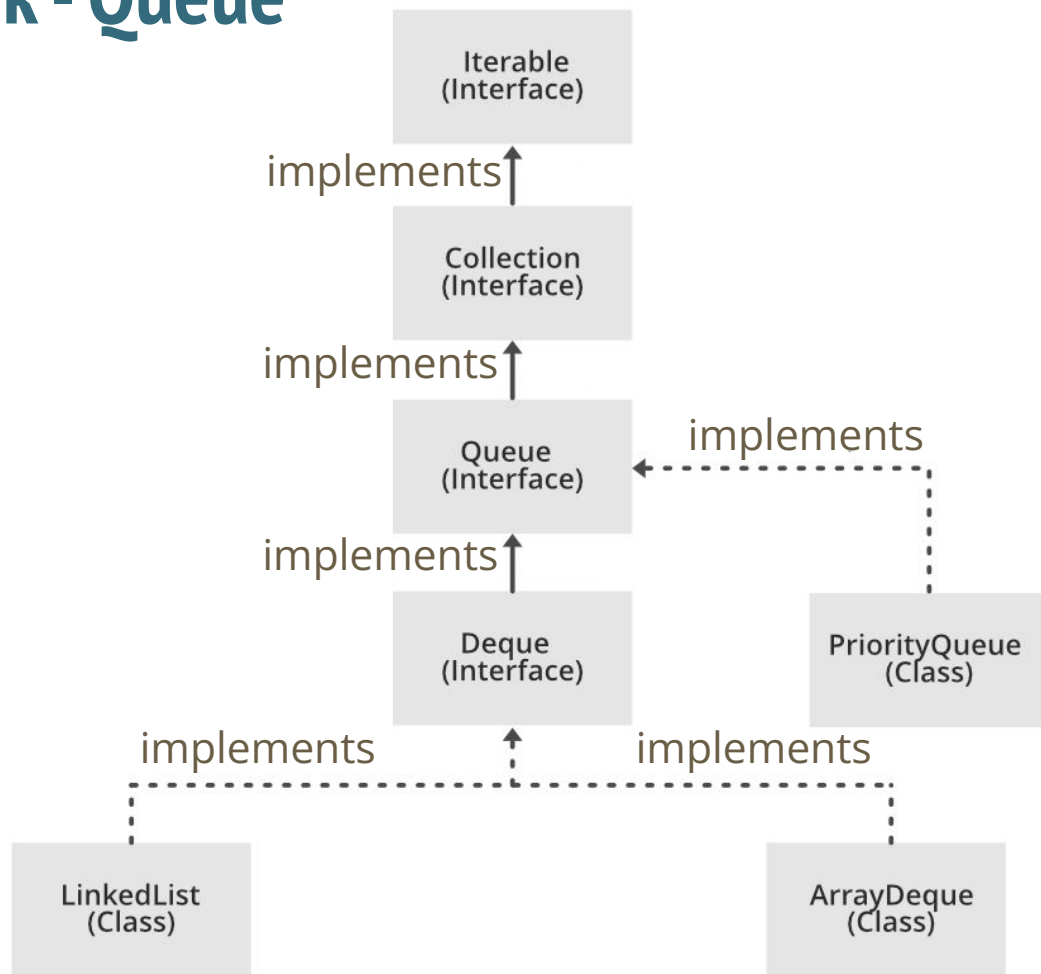
Collections Framework - Set



Collections Framework - Set

Feature / Class	HashSet	LinkedHashSet	TreeSet
Duplicates Allowed	No	No	No
Ordering	No ordering (hash-based)	Insertion order maintained	Sorted order (natural/comparator)
Implementation	Hash table internally	Hash table + linked list	Balanced tree (Red-Black Tree)
Performance	$O(1)$ average for basic ops	Slightly slower due to ordering	$O(\log n)$ for basic ops

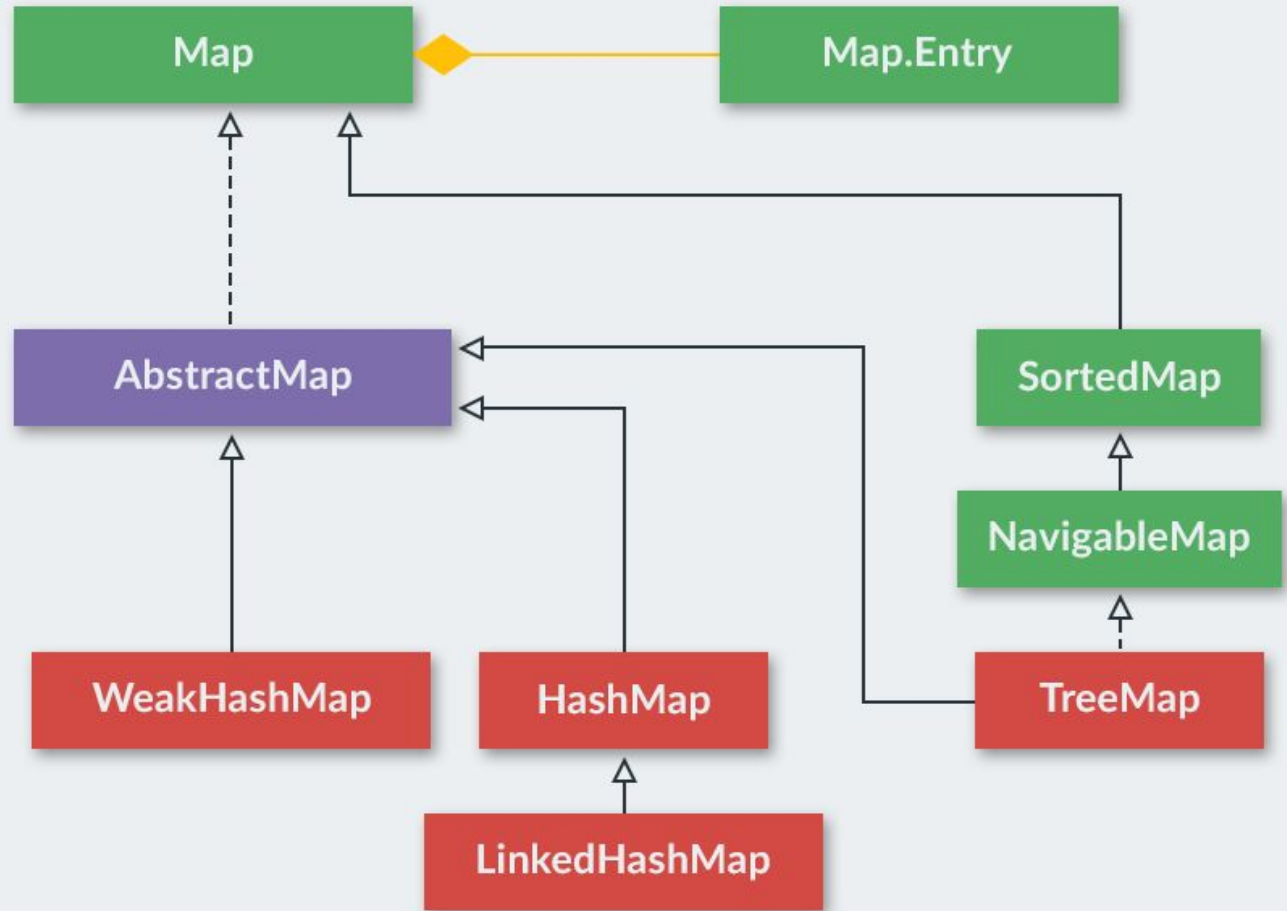
Collections Framework - Queue



Collections Framework - Queue

Feature / Class	PriorityQueue	LinkedList	ArrayDeque
Interface Implemented	Queue	List, Queue, Deque	Deque
Ordering	Priority-based ordering	FIFO (if used as queue)	FIFO (queue) or LIFO (stack)
Data Structure	Heap	Doubly linked list	Resizable array
Null Elements Allowed	No	Yes	No

Collections Framework - Map



interface abstract class class extends implements

Collections Framework - Map

Feature / Class	HashMap	Hashtable	LinkedHashMap	TreeMap
Key Ordering	No ordering	No ordering	Order of insertion	Sorted (natural/comparator)
Null Keys	Allows one null key	Does not allow null keys	Allows one null key	Does not allow null keys
Thread Safety	Not synchronized	Synchronized (Thread-safe)	Not synchronized	Not synchronized
Performance	Generally $O(1)$ for get/put	Slower due to synchronization	Slightly slower than HashMap	$O(\log n)$ due to tree structure



Thank You

Accelerate your evolution – and stay a step ahead!